

PROFESSIONAL SUMMARY

Final year AI & ML engineer with internship experience in AI, ML, Data Science and Deep Learning, skilled in Python, machine learning, and deploying deep learning models. Proven through internships, SIH 2025 Finals and Top-4 finish at IIT Bombay CTD International Hackathon.

SKILLS

Languages

Python, Java, JavaScript, SQL

AI/ML & Data

Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG, TensorFlow, PyTorch, OpenCV, Scikit-learn, ONNX

Data & Stats

Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis

Frameworks

Flask, FastAPI, Node.js, Fast MCP

Databases

MongoDB (Certified), MySQL, PostgreSQL, SQLite, ChromaDB

Tools & Platforms

Docker, REST APIs, Render, Railway, Firebase, Git/GitHub

Soft Skills

Problem-Solving, Teamwork, Communication, Agile Collaboration

CERTIFICATIONS & AWARDS

Fundamentals of MCP, Hugging Face, January 2026

MongoDB Node.js Associate Developer, MongoDB University, December 1st, 2024

Generative AI with Diffusion Models, NVIDIA, September 1st, 2025

Create Your First Gemini Enterprise Application, Google, May 2026

SIH 2025 National Finalist, Smart India Hackathon, December 2025

Top 4 Finalist, CTAI International Hackathon, IIT Bombay, October 2025

Finalist — AI Summit 2026 Buildathon (HCL & GUVI), HCL/GUVI, February 2026

AREAS OF INTEREST

- Machine Learning & Deep Learning
- Natural Language Processing
- Large Language Models
- Retrieval-Augmented Generation
- Agentic AI & AI Systems

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning, Kongu Engineering College, Erode May,2027

- CGPA: 8.49

EXPERIENCE

AI / Data Science Intern, Twincord Technologies

July 2025 – October 2025

- Deployed FastAPI ML inference endpoints; monitored latency and optimized scalability of AI platform.
- Collaborated with senior developers on algorithm implementation, model validation, and troubleshooting.

Deep Learning Intern, ShineLogics

March 2025 – May 2025

- Trained CNN models (ResNet, EfficientNet, MobileNet) on GPU for real-time product identification; deployed and validated on edge hardware over 2-week production trial with 98%+ uptime and zero critical failures.
- Optimized model inference pipeline reducing latency by 35%; tested across 70+ product images during real-time deployment validation.

Backend Developer Intern, ZEONE

November 2024 – January 2025

- Built 6 RESTful APIs in Node.js/Express for an AI-powered interview application; optimized database queries and caching reducing average API response time from 1.2s to 450ms (62% improvement).
- Implemented JWT authentication and request validation; processed 500+ concurrent interview sessions during QA testing with zero downtime.

PROJECTS

ProTube – Distraction-Free Educational Video Platform,

Tech: React, Node.js/Express, Supabase, Redis, TailwindCSS, JWT

August 2025 (github.com/vijayakanth06/protube)

- Eliminated video distractions by architecting a long-form streaming platform with short-form content blocking (99.2% accuracy), AI educational search, and a Chrome curator extension featuring O(1) duplicate detection reducing curation time from 15 min to <30 sec.
- Tested with 40+ users; achieved 2.8s average page load time and 94% user retention across 3-week pilot phase.

Fabric Defect Detection System - SIH 2025 Finals,

Tech: Python, TensorFlow, CNN, OpenCV, Flask, IOT Integration

December 2025 (github.com/vijayakanth06/Fabric_defect_detect)

- Automated industrial quality inspection by deploying a real-time CNN defect detection pipeline (ResNet/EfficientNet) achieving 96.8% accuracy on 2,000+ fabric samples; integrated edge IoT hardware for production floor deployment.
- Secured SIH 2025 Finals qualification; demonstrated live defect detection with <200ms inference latency and zero false negatives on critical defects across 50+ test runs.

Fabcluster – Spectrophotometer QC & Color Shade Clustering

Analyzer, Tech: Python, Streamlit, SciPy, NumPy, Pandas, Plotly, SQLite

February 2026 (github.com/vijayakanth06/spectrophotometer)

- Eliminated fabric shade misclassifications in textile dyehouses by developing a complete-linkage hierarchical clustering engine using CIE L*a*b* color science and 4-component confidence scoring; achieved 97.3% clustering precision across 2,000+ fabric samples.
- Automated PDF lay-plan generation reducing QC report time from 45 min to 3 min (93% reduction); deployed as Streamlit app serving 15+ textile mills during pilot phase.

ResumeForge- MCP-Based NLP Resume Pipeline,

Tech: Python, MCP agents, NLP, RAG, FastMCP, LLM

April 2026 (github.com/vijayakanth06/ResumeForge)

- Achieved 87% ATS parsing accuracy by building an open-source 5-tool FastMCP server aggregating LinkedIn, GitHub, and LeetCode data into clean Markdown; cutting resume tailoring time from 2 hours to <10 seconds (99% time reduction).
- Processed 200+ resume variants during testing; averaged 4.2s end-to-end pipeline execution with zero data loss across LinkedIn-to-ATS workflows.